1. Title, option number, team members (names and NetIDs) and who does what.
    a. RTL Doctor: an agent that finds. explains, and repair bugs in Verilog designs
    b. Backup option: Option 5 (battery-health)

Name	NetID	Does
Pranav Senthilkumar	pranav_s79	bug-detection model (fine-tune CodeT5+), dataset labeling
Krish Pokle	krishpokle	Repair model (deepseek-coder), tools (Yosys/Icarus), portal

2. Three questions that a user will ask your agent.
    a. Can you detect any bugs in my Verilog RTL design?
    b. Can you explain this bug and fix it?
    c. Can you visualize my RTL design as a gate level schematic?
3. Data: the link, the size, and what one row (or image, or record) is, with a df.head() and one plot.
    a. https://huggingface.co/datasets/shailja/Verilog_GitHub
    b. 108,971 rows
    c. module_code | label (clean/buggy) | bug_type | bug_line | original_code
    d. https://colab.research.google.com/drive/1z6t8CuhmoA8v80w-gNCeGrBM6080TG1N?usp=sharing
4. Models: for each one, inputs, output, method, metric, and the baseline it must beat (for example, always predicting the most common class).

	Bug detector	**Repair Model**
Input	Verilog Module	Buggy module + bug type
Output	Clean/buggy + bug type	Patched Verilog module
Method	Fine-tine codeT5 + 220M encoder with a classification head	LoRA fine-tine deepseek-coder-1.3B-instruct on pairs
Metric	Macro-F1 over bug types	Fix rate: & of patches that compile and match the original's behavior
Baseline to beat	Always predicting the most common class	Return the input unchanged

5. Tools: the name, inputs and output of each one, one line each.

    a. lint_rtl(code) → list of warnings
    b. detect_bugs(code) → CodeT5+ model
    c. repair_rtl(code, bug_type) → deepseek-coder model
    d. simulate(code, testbench) → Icarus Verilog
    e. render_schematic(code, testbench) → Yosys synth → netlistsvf
    f. send_report(to_email, summary) → sent/failed
6. The portal: a sketch of the chat and the dashboard (a photo of a drawing is fine), the charts it will show, and what the report email will say.
    a. Left: Will show a chat, allow you to upload a .v file, ask questions, and see answers with code diffs.
    b. Right: Will show a dashboard, bar chart of bugs found by type, per-line confidence, before/after repair, and schematic viewer
7. Guardrails that you plan, from the menu, and the layer of each.

Guardrail	**Layer**
Accept only .v/.sv files, size limit	Input
Refuse non-RTL questions	Input/prompt
Prompt-injection filter	Input
Sandbox + timeout on Yosys/Icarus runs	Tool/execution
Never show a “fix” that doesn’t compile	Output
User confirms before an email is sent	Action

8. The LLM: TAMU AI (which model) or another one, and the account that will send the report email.
    a. TAMU AI (5.6 sol)
    b. rtldoctor@gmail.com
9. What will be working by Fri, Nov 20.
    a. dataset built with the bug injection
    b. code T5+ detector trained, F1 reported against both baselines
    c. lint_rtl and render_schematic tools working
    d. Chat answers Q1 and Q3 end to end
    e. Repair model: first LoRA run with fix-rate number; dashboard and email may still be partial